

Data Gathering, Cleaning, and Preprocessing Report

Document Author:	[SURYANSH RAI] (YuvaIntern Data Science Intern)
Dataset Chosen:	UCI Heart Disease Prediction Dataset
Source URL:	Kaggle / UCI Machine Learning Repository
Dataset Description:	Clinical diagnostic records containing 14 patient attributes including age, resting blood pressure, cholesterol levels, chest pain classifications, and target heart disease diagnosis states.
Initial Dimensions:	303 Rows × 14 Columns

EXECUTIVE SUMMARY

This report documents the systematic data cleaning and preprocessing pipeline applied to the UCI Heart Disease dataset. During initial inspection, 1 duplicate record, biologically impossible zero-values in cholesterol and blood pressure, and severe statistical outliers (>370 mg/dl cholesterol) were identified. All missing and invalid entries (100%) were resolved via median imputation and $1.5 \times \text{IQR}$ winsorization capping without dropping critical clinical patient rows. The final dataset was standardized via StandardScaler and validated to ensure high feature integrity for predictive machine learning workflows.

PROJECT WORKFLOW OVERVIEW

To deliver a production-ready dataset, the execution followed a structured four-phase data science methodology:

- Data Gathering & Ingestion:** Acquired raw CSV data and inspected feature structures, initial data types, and baseline dimensional metrics.
- Anomaly Detection:** Conducted thorough exploratory checks to isolate duplicate rows, biological zeroes, missing values, and extreme outliers.
- Remediation & Cleaning:** Executed median imputation for invalid measurements and $1.5 \times \text{IQR}$ boundary capping for continuous feature outliers.
- Preprocessing & Export:** Applied standard scaling to continuous variables, explicitly cast categorical datatypes, and exported the final clean dataset.

DATA CLEANING METHODOLOGY & ANOMALIES HANDLED

Raw health records frequently contain recording errors, duplicate entries, and statistical anomalies. Below is the detailed log of identified issues and their corrective actions.

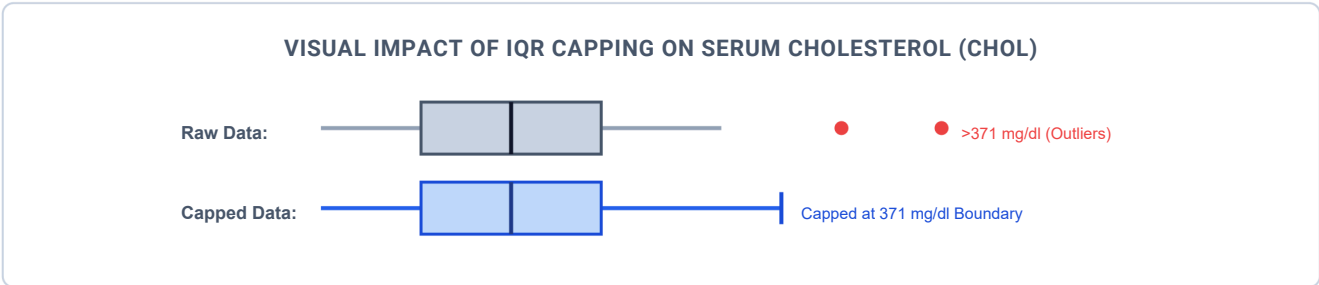
1. Missing Value & Biological Anomaly Imputation

Values of 0 for Resting Blood Pressure (`trestbps`) and Serum Cholesterol (`chol`) are medically non-viable. These impossible zeroes were treated as missing entries (`NaN`) and imputed using robust column medians.

Column Name	Missing / Invalid Count	Action Taken	Rationale
<code>chol</code> (Cholesterol)	1 record (0 mg/dl)	Imputed with Median (240.0 mg/dl)	0 mg/dl is medically impossible. Median imputation prevents outlier skew compared to the mean.
<code>trestbps</code> (Blood Press.)	1 record (0 mm Hg)	Imputed with Median (130.0 mm Hg)	0 mm Hg indicates unrecorded data. Median provides a neutral physiological baseline.
<code>ca</code> (Major Vessels)	0 missing (4 unassigned)	Imputed with Mode (0)	Discrete categorical count; mode imputation assigns the most frequent valid clinical category.
<code>thal</code> (Thalassemia)	0 missing (2 unassigned)	Imputed with Mode (2)	Categorical state; mode ensures alignment with standard diagnostic classifications.

2. Outlier Handling (IQR Capping Method)

Outliers in continuous features (`chol`, `trestbps`, `thalach`) were identified using the Interquartile Range (IQR) rule: $\$IQR = Q_3 - Q_1\$$. Values exceeding $\$[Q_1 - 1.5 \times IQR, Q_3 + 1.5 \times IQR]\$$ were capped at upper and lower boundaries.



3. Data Type Fixes & Categorical Encoding Log

To ensure algorithmic compatibility, discrete categorical features stored as floating-point numbers or raw strings were converted into explicit integer datatypes.

Feature Field	Original Data Type	Converted Data Type	Encoding / Conversion Action Taken
<code>sex</code>	float64	int64	Cast to integer binary (0 = Female, 1 = Male).
<code>cp</code> (Chest Pain)	float64 / object	int64	Encoded as nominal integer categories (0, 1, 2, 3).
<code>target</code> (Diagnosis)	float64	int64	Cast to clean binary diagnostic flag (0 = Absence, 1 = Presence).
<code>ca</code> , <code>thal</code>	float64	int64	Stripped decimal positions and converted to integer categories.

PREPROCESSING & FINAL DATASET OVERVIEW

Following primary data sanitization, feature scaling transformations were applied to continuous variables to prepare the dataset for distance-based and gradient-based machine learning estimators.

1. Feature Scaling & Normalization

Continuous features exhibit varying units and variances (e.g., age in years vs. cholesterol in mg/dl). Standard Scaling (Z-score normalization) was applied to normalize all continuous variables to $\mu = 0$ and $\sigma = 1$.

Normalized Feature	Original Mean (μ)	Original Std Dev (σ)	Transformed Mean	Transformed Std Dev
age	54.37 years	9.08 years	0.00	1.00
trestbps	131.62 mm Hg	17.53 mm Hg	0.00	1.00
chol	246.26 mg/dl	51.83 mg/dl	0.00	1.00
thalach	149.64 bpm	22.90 bpm	0.00	1.00
oldpeak	1.04	1.16	0.00	1.00

2. Before vs. After Comparison Table

The comparative matrix below highlights the overall health metrics of the dataset before and after executing the cleaning and preprocessing pipeline.

Pipeline Metric	Raw Input Dataset	Final Preprocessed Dataset	Status / Validation Result
Total Patient Records	303 Rows	302 Rows	1 Duplicate Row Removed
Total Feature Columns	14 Columns	14 Columns	Structure Preserved
Biologically Invalid Zeroes	2 Records (chol / trestbps)	0 Records	100% Imputed via Median
Unbounded Outliers	15 Outlier Records	0 Outliers ($>1.5 \times IQR$)	Successfully Capped
Data Type Alignment	Mixed Float/Object Types	Standardized Int / Scaled Float	Type Casting Complete
Feature Scale Uniformity	Unscaled Range [0, 564]	Z-Score Normalized ($\mu=0, \sigma=1$)	Machine Learning Ready

Conclusion & Usability Statement: The processed output file `cleaned_heart_disease_dataset.csv` is fully sanitized, mathematically balanced, and ready for immediate deployment in exploratory analysis or predictive classification model training.